

Week 5 — Effect modification and interaction: which question, which adjustment set, which scale

Duration: ~2.5 h (Tue 6 Oct, before the Thu 8 Oct lab). Deck: `lectures/slides/Week5_interaction_slides.qmd`. Its numbers come from `lectures/analysis/Week5_joint_exposure.R` (written to `Week5_joint_exposure.json`), and from the P2, P3 and M1 result objects (`examples/_results/`). The code and output on its twelve “In R” slides come from `lectures/analysis/Week5_code_outputs.R`, which runs each snippet exactly as shown and records what R printed.

Timing. Per-slide markers below sum to **132 min including the break**, against a 150-minute cap: 18 minutes of slack. The deck’s speaker notes name the plan slide for every deck slide and carry the same minutes; Plan Slide 8 spans two deck slides, and the “In R” slides are lettered (9b–c, 10a, 12a, 15a, 16a–c, 20a, 22a, 23a). **If you retime a slide, retime it in the deck’s notes too.** See *If you are running behind* at the foot of this file.

Built on Week 4, on purpose. Week 4 taught confounding with **obesity alone** and handed this lecture the paper’s real exposure: “*Next: Week 5 — the paper’s four BMI $\times$ waist groups*”. Week 5 keeps Week 4’s rows (**5,911**), its **pre-exposure adjustment set** (age, sex, race/ethnicity, smoking, sedentary time), its DAG conventions, its notation (A, Y, L, U; potential outcomes), its “Read more” lines, and four of its results as callbacks: the collapsibility table (read for modification), the g-computation idea (extended to standardised survival), the Table 2 DAG (Q1 and Q2 are today’s two readings), and the Q2 estimate for waist fat, **1.24 (0.93–1.67)**, which today splits by obesity.

Concrete after the break. From the scale slides onward every idea is followed by an “In R” slide: the code that computes it, on our data or the toy tables, with R’s real output underneath (reprex style, `#>`). Students see where each number comes from before they see it summarised. Nothing on those slides is typed: the code runs as shown, and a check fails if a slide’s block differs from what R printed.

The lecture is one argument. **The paper’s four phenotype groups are two exposures crossed, and the 2×2 asks two different questions.** *Effect modification* — does waist fat’s effect on death differ between the obese and the not obese? — needs only waist fat’s confounders. *Interaction* — what do obesity and waist fat do together? — needs the confounders of **both**, and obesity’s include serious illness and earlier disease, which nobody measured. So in MASLD the first question is answerable only under strong, stated assumptions, and the second is not answerable even under those. Then the scale: the answer depends on whether you compare ratios or differences, and the additive scale needs the lowest-risk cell as its reference — which here is Group I, with **4 deaths**. Counting the cells first is the habit the lecture exists to build, and the sex example shows what happens when nobody does.

Every DAG is drawn. One new figure, `lectures/img/w5_two_questions.png`, in Week 4’s visual language (colour = role; dashed outline = not measured; square = conditioned on; grey arrow with “?” = the effect we want; dashed red band = an open back-door path). Every adjustment-set claim, including each broken assumption on the second Slide 8, is checked with dagitty, and the deck prints the DAG as dagitty code in an appendix so students can check it themselves. Week 4’s `w4_table2_panels_a.png` is reused on Plan Slide 9. The two data figures (`w5_four_groups.png`, `w5_em_forest.png`) read every number from the emitted JSON rather than having it typed in; the four-group figure draws comparisons as lines without arrowheads, so they cannot be read as causes.

Row-set discipline. Every quantitative slide names its rows. Six sets appear: - **5,911** — Week 4’s rows: the locked domain restricted to people complete on every covariate of the paper’s Model 2 and of the pre-exposure set. All of Part 4 (each slide says so in its footer). - **6,048** — the locked domain itself. Only the retired crude IV-vs-I table by sex (Plan Slide 22), on the rows P3’s code uses. Group I has 375 people there, not 355. - **1,296** — P3’s rows: Groups III + IV complete on Model 1 (Plan Slide 23, P3’s numbers). - **1,291** — Groups III + IV within the 5,911 (Plan

Slide 23, the pre-exposure row). - **5,939** — P2’s Model-1 complete cases, for the 1.06 quoted in Plan Slide 18’s notes. - **6,371** — the full file, only in the provenance note on Plan Slide 22.

What this week does NOT do. Survey weights are Week 6: every number here is **unweighted**, and the design-aware sex ratio (M1) is quoted only as the hand-off. Propensity-score methods for subgroup effects (Karim, ch. 8, §8.6) come after the propensity-score weeks. Missing data is Week 7.

Learning objectives

1. Distinguish **effect modification** (one exposure; its effect varies across levels of M) from **interaction** (the joint effect of two exposures), and from confounding and from a bare product term.
2. Choose the **adjustment set** each question needs, using a DAG: A’s confounders for effect modification; both exposures’ confounders for interaction — and state the assumptions the DAG makes.
3. Explain why effect modification is **scale-dependent**, compute RERI, AP and S, and know that all three are additive-scale measures.
4. **Count** people and outcomes per cell before trusting an interaction measure, and choose the lowest-risk cell as the reference.
5. Use the right **time-to-event** tools: a Cox product term (multiplicative) and standardised risks at a fixed time (additive).
6. Report a subgroup finding as evidence, not mechanism.
7. Write the R for each of these: stratum-specific HRs from a product term, the likelihood-ratio test, standardised risks, and RERI / AP / S / IC.

Continuity

- **Last week (Week 4 / L3).** Confounding, with obesity as the only exposure. Crude HR 0.67 moved to 0.95 with the pre-exposure set; the DAG decided what to adjust; collapsibility showed ORs and HRs move without confounding; Table 2 showed one model answers one question, and gave waist fat’s own answer as 1.24 (0.93–1.67).
 - **Today.** The paper’s real exposure, obesity $\times$ waist fat. Same rows, same adjustment set, same figure style.
 - **Thursday (L4).** EpiMethods **confoundingE2**: the **RHC** data again, a **binary** outcome, DNR as the modifier, logistic regression, RERI/AP/S. See *Thursday* below.
 - **P3.** One pre-specified modifier on the student’s own paper (optional, ungraded; enters M1).
-

Opening · 9 minutes

Slide 1 — Does waist fat matter more for some people? (4 min)

- The four groups as a 2×2 , with people and deaths: **I 355 / 4 · II 4,265 / 362 · III 732 / 67 · IV 559 / 101** (5,911 rows).
- Week 4 asked about obesity alone. Today: does waist fat’s effect depend on obesity, and what do the two do together?
- Point at Group I’s **4 deaths** now. The lecture comes back to it four times.

Slide 2 — Our running example (3 min)

- A = high waist fat ($\text{WhtR} \geq 0.6$); M = obesity ($\text{BMI} \geq 30, \geq 25$ for Asian participants); Y = death; L = Week 4’s pre-exposure set.
- **Why not Model 1?** Week 4 showed some of its conditions are plausibly mediators, and reported the pre-exposure set as “the closest measured set to the DAG’s”.
- **Income** was in Week 4’s L but is missing for 516 of the 5,911; it returns as an assumption on Slide 8.
- The paper compares every group with Group I and never asks either question.

Slide 3 — Objectives (2 min)

Part 1 · Two questions that sound alike · 14 minutes

Slide 4 — Effect modification: one exposure (5 min)

- Definitions on both scales, with potential outcomes conditioned on M (Karim ch. 8, §8.4): $RD_m = P[Y(1) = 1 \mid M = m] - P[Y(0) = 1 \mid M = m]$; RR_m likewise.
- **M's own effect needs no causal story.** The exposure needs exchangeability, positivity and consistency (Week 4); the modifier does not (Karim §8.4.1). Sex can modify an effect; nobody can set it. **But M must be something A cannot change** (Slide 9a shows why).

Slide 5 — Interaction: two exposures (5 min)

- Four potential outcomes $Y(a, m)$. Synergy / antagonism. The deck keeps the letter M for the second variable throughout, as Karim does; EpiMethods writes B.
- Interaction is **symmetric**; effect modification is not — “waist fat by obesity” and “obesity by waist fat” are different questions (Slide 9).

Slide 6 — Confounding, modification, interaction (4 min)

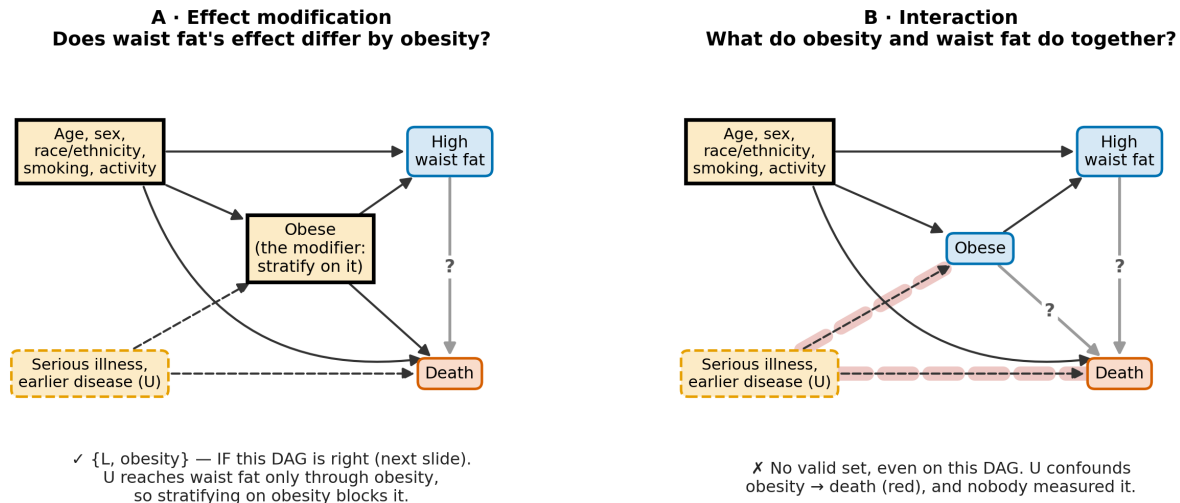
- Remove confounding; report modification; report all four combinations for interaction. A product term is a model device serving both — if the model has that question's adjustment set.
- The same variable can be both a confounder and a modifier: obesity confounds waist fat $\rightarrow$ death (Week 4, Table 2 Q2) and is today's modifier.

Part 2 · What to adjust for · 26 minutes

Slide 7 — Each question has its own adjustment set (4 min)

- EM of A by M: confounders of $A \rightarrow Y$. Interaction of A and M: confounders of $A \rightarrow Y$ **and** $M \rightarrow Y$ (EpiMethods *Concepts*, “Implications for Confounding Control”).
- **The EpiMethods effect-modification slides** run the classic example. Income as a modifier of smoking $\rightarrow$ hypertension: {age, gender}. Income as a second exposure: add diet or education — and the slides ask “but how?”, because income has no arrow into hypertension (a real exposure, or a proxy for education?). Their “true” interaction is education as the second exposure, needing only {age, gender}. Today asks the same of obesity: a second exposure, or only a sorter?

Slide 8 — In MASLD, one question needs less (10 min) · two deck slides · do not drop



Square = conditioned on (adjusted, stratified, restricted or selected). Dashed outline = not measured. Grey arrow with ? = the effect we want.

First deck slide — the figure.

- **Panel A — effect modification.** Waist fat $\leftarrow$ obese $\leftarrow$ U $\rightarrow$ death passes through obesity; stratifying on obesity blocks it. **{L, obesity}** is the one minimal set **on this DAG**.
- **Panel B — interaction.** Obesity is now an exposure; obese $\leftarrow$ U $\rightarrow$ death is open. No measured set closes it. With U measured, **{L, U}** would do.
- **U merges Week 4's two latent nodes** — serious illness, and disease that came before obesity. Week 4's appendix needed {Before, Ill, L} for obesity's effect for the same reason.

Second deck slide — “What Panel A's ✓ assumes”. Each is dagitty-checked: break it and Panel A's set disappears.

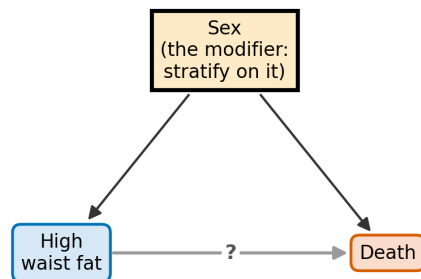
1. **Obesity comes before waist fat** (Week 4's Table 2 DAG; Week 4 flagged the arrow: “the DAG, not the data, decides”). If waist fat drives BMI, obesity is caused by the exposure and cannot be the modifier.
 2. **U reaches waist fat only by moving people across BMI 30.** A strong assumption: illness can shrink the waist without changing the obesity category. Sensitivity: adding BMI itself to the model gives **2.80** and **1.07** (ratio **2.62**), against 2.99 and 1.09.
 3. **Nothing unmeasured causes both obesity and waist fat** (a fat-distribution gene, say). Obesity is a collider of U and such a cause, so stratifying on it would open waist $\leftarrow$ G $\rightarrow$ obese $\leftarrow$ U $\rightarrow$ death.
 4. **L is complete** — Week 4's L also had income, missing for 516 of the 5,911.
- **And under all of it, the FLI ≥ 60 selection.** FLI is computed from BMI **and** waist, so the cohort is selected on a child of both exposures. That biases even Panel A; Week 4 named the direction for IV vs III — downward (the not obese with lower waist fat reach FLI ≥ 60 with higher triglycerides and GGT).
 - **The honest summary:** modification is answerable only under these assumptions; interaction is not answerable even under them. In this DAG, the interaction question needs everything the modification question needs, plus obesity's confounders.

Slide 9 — Which way round? Choosing the modifier (3 min)

- Reuses Week 4's Table 2 panels (Q1 obesity, Q2 waist fat). Waist fat by obesity: obesity is a confounder → stratifying helps. Obesity by waist fat: waist fat is a mediator → stratifying removes part of obesity's effect. The Week 4 figure was drawn without U; with it (Slide 8) obesity's own effect has no valid set either, so this reading fails twice.
- Which variable comes first decides which reading works. The next slide makes that a rule.

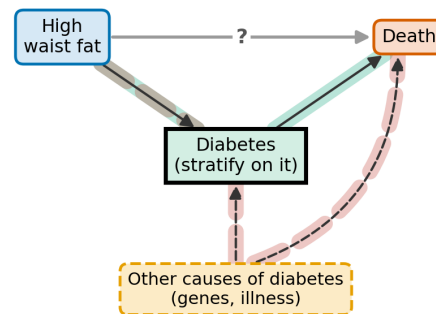
Slide 9a — The exposure must not change the modifier (5 min)

A · Fine: the exposure cannot change it



Waist fat cannot change anyone's sex: each stratum holds the same people, whatever their waist size.

B · Not fine: the exposure can change it



Stratifying on diabetes (1) blocks part of waist fat's effect (green) and (2) opens waist fat → diabetes ← other causes → death (red).

Square = stratified on. Dashed outline = not measured. Grey ? = the effect we want. Green = part of the effect; red = an open non-causal path.

- The rule, stated so students can apply it:** *the modifier must be something the exposure cannot change.* “Waist fat's effect among people with $M = m$ ” asks what changing waist fat would do **to the people in that group**; that only makes sense if changing waist fat cannot move people into or out of the group.
- The safe check is time:** sex, age, genes, a condition that began **before** the exposure. *Recorded at the same visit* is not enough — ask whether the exposure could have caused it. Diabetes at the NHANES exam fails; obesity passes only if it came before the waist fat. (Timing is the check, not the rule: something that arises later but cannot be affected by the exposure, such as calendar time, is also fine.)
- Panel A (fine): sex.** Waist fat cannot change anyone's sex, so each stratum holds the same people whatever their waist size. The modifier may also cause the exposure or be a confounder; obesity is allowed, but only under the arrow obese → waist fat.
- Panel B (not fine): diabetes,** plausibly caused by waist fat. Stratifying on it does two things at once (both dagitty-checked):
 - it **blocks part of the effect** — the route waist fat → diabetes → death;
 - it **compares unlike people** — among people with diabetes, those with lower waist fat got it some other way (genes, illness) that also raises the risk of death: waist fat → diabetes ← other causes → death is opened. That is collider bias, Week 4's FLI lesson in subgroup form.
- If you care about a downstream variable,** the question is **mediation:** the effect with diabetes held fixed (a controlled direct effect, which needs diabetes' own confounders measured), or the effect among people whose diabetes would not change with waist fat (a principal stratum — not a subgroup you can see in the data).
- The “obesity paradox” has panel B's shape in heart disease (Banack & Kaufman, *Prev Med* 2014) and in diabetes (Lajous et al., *Epidemiology* 2014); how much of it collider bias explains is debated.

Timing unknown (Thursday’s DNR order and catheterisation, both day 1)? State the assumption.

Slides 9b–9c — In R: could waist fat cause the modifier? · what splitting by diabetes does (2 + 2 min)

- 9b: the diabetes $\times$ waist-fat table and its column percentages (13% vs 28%).
- 9c: one Cox model per diabetes stratum (Karim ch. 8, §8.4.3.2), adjusted for obesity and L. High vs lower waist fat: **1.23 (0.85–1.78)** without diabetes, **0.93 (0.56–1.54)** with it — a gap well within noise. Diabetes is twice as common with high waist fat (28% vs 13%).
- **Neither** number is waist fat’s effect in a fixed group: waist fat moves people into one stratum and out of the other, so both strata cut the diabetes route and may compare unlike people. 1.23 being close to Week 4’s 1.24 does not rescue it.
- The slide’s footer carries the variable key and the script path; the “In R” slides are one R session, run in order.

Part 3 · Which scale? · 19 minutes

Slide 10 — Last week’s table, read for modification (3 min)

- Week 4’s collapsibility toy: OR 4.00 in both age groups, RD 0.30 in both, RR 1.60 vs 2.50. Age modifies the RR and neither the OR nor the RD. Hence **effect-measure modification**.
- Week 4’s notes promised this callback.

Slide 10a — In R: the same table, three scales (2 min)

- The toy table as a data frame; one line each for RD, RR and OR. The RR differs by age (1.6 vs 2.5), the OR (4) and the RD (0.3) do not.

Slide 11 — Two scales, two questions (6 min)

- **Every term is defined on the slide:** R_{am} (risk with $A = a$, $M = m$), $RR_{am} = R_{am} \div R_{00}$, IC (interaction contrast), RERI (relative excess risk due to interaction), AP (attributable proportion), S (synergy index).
- Multiplicative: ratio of HRs = $\exp(\text{product-term coefficient})$; 1 = none.
- Additive: $RD_1 - RD_0$; $IC = R_{11} - R_{10} - R_{01} + R_{00}$; $RERI = RR_{11} - RR_{10} - RR_{01} + 1$ (0 = none); $AP = RERI / RR_{11}$ (0 = none); $S = (RR_{11} - 1) / [(RR_{10} - 1) + (RR_{01} - 1)]$ (1 = none). **All three are additive-scale measures, and RERI and AP can be negative with no lower bound.**
- RERI is defined relative to the doubly unexposed cell, so it needs that cell to be the **lowest-risk** one; recode a protective exposure first (Knol et al. 2011). Thursday’s lab does that recoding.
- **Pre-empt the lab text.** `confoundingE2` describes S as “greater than expected based on multiplicative effects” and “SI = 1: no multiplicative interaction”. S is an additive measure. Say so here so the lab does not undo it.

Slide 12 — Same data, two answers (3 min)

- Made-up 2×2 : 5%, 10%, 15%, 30%. RR 2, 3, 6 = $2 \times 3 \rightarrow$ no multiplicative interaction; RERI = 2, IC = +10 points $\rightarrow$ positive additive interaction.
- When both exposures raise risk, no multiplicative interaction implies positive additive interaction: $RERI = (RR_{10} - 1)(RR_{01} - 1) > 0$. Report both scales, and say which scale each verdict refers to.

Slide 12a — In R: RERI, AP, S and IC from four risks (3 min)

- The same four risks in R: ratio 1 (no multiplicative interaction); RERI 2, AP 0.33, S 1.67, IC +0.10 (positive additive interaction). Part 4 runs exactly these formulas on hazard ratios and standardised risks.

Slide 13 — What to report (2 min)

- Knol & VanderWeele (2012), as the EpiMethods reporting table lays it out: cell counts with outcomes; every combination vs one reference; A within M (EM: this subset); M within A (interaction only); both scales with CIs; the adjustment set.

Slide 14 — BREAK (10 min)

Part 4 · The four groups, analysed · 33 minutes

Slide 15 — Count first (3 min)

- The 2×2 again, as counts. Every comparison with Group I divides by **4 deaths**, and Group I is the **lowest-risk** cell, so it is the reference RERI needs.
- **Week 4's positivity check, for the new exposure.** Of 2,432 obese women, only **57** have lower waist fat; among the obese, the chance of high waist fat given L runs from 27%–99.7% (3%–95% among the not obese). For obese women, the obese-stratum risks lean on the model more than on data.

Slide 15a — In R: count first (2 min)

- `table(group, died)` and the obese-women count: two lines before any model.

Slide 16 — One model, two ways to write it (4 min) · now with its R output

- `group + L` and `obese * central + L` are the same fit: the slide runs both and prints `all.equal(...)` = TRUE, then the three coefficients: `obese` 0.34 (obesity's HR among those with lower waist fat, = $1 \div 2.95$), `central` 1.09, `obese:central` 2.74. EpiMethods `confounding9` shows the same equivalence on NHANES (obesity $\times$ race/ethnicity, hypertension).
- In the product form, `central` is waist fat's effect among the not obese **only**. Reading it as “the effect of waist fat” is the Table 2 fallacy for a modifier's coefficient (Westreich & Greenland 2013).
- `interactionR`'s effect-modification table treats the **first** exposure named as the modifier — check which way round it prints (Thursday).

Slide 16a — In R: which hazard ratio is which (3 min)

- Add coefficients for each stratum's HR (1.09; 2.99). For a CI, **flip the coding** so the other stratum is the reference (Karim ch. 8, §8.4.3.1): `central` in the flipped model is waist fat among the obese, 2.99 (1.11–8.09).

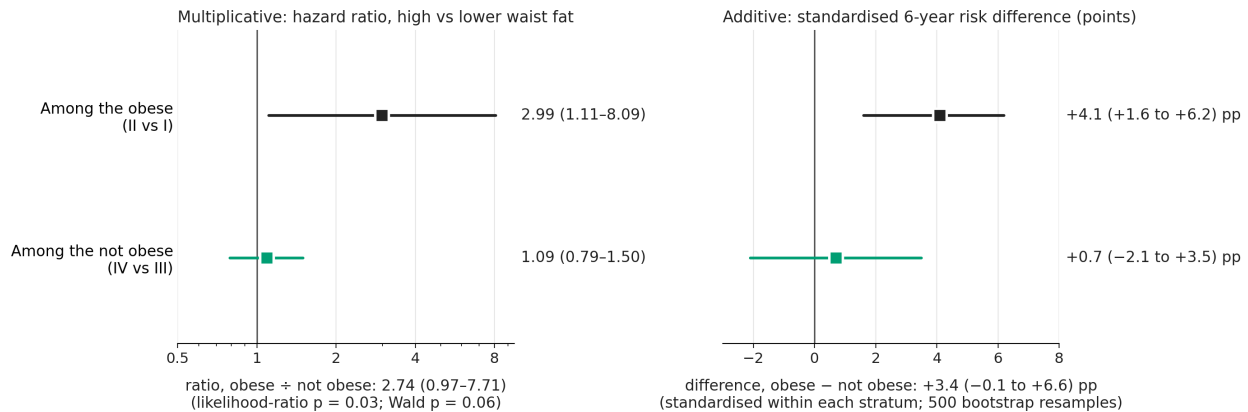
Slide 16b — In R: does the product term matter? (2 min)

- `fit_main` (no product term) is Week 4's Q2 model: waist fat 1.24. `anova(fit_main, fit_prod):` ² 4.75, likelihood-ratio p 0.029.

Slide 16c — In R: standardised six-year risks (3 min)

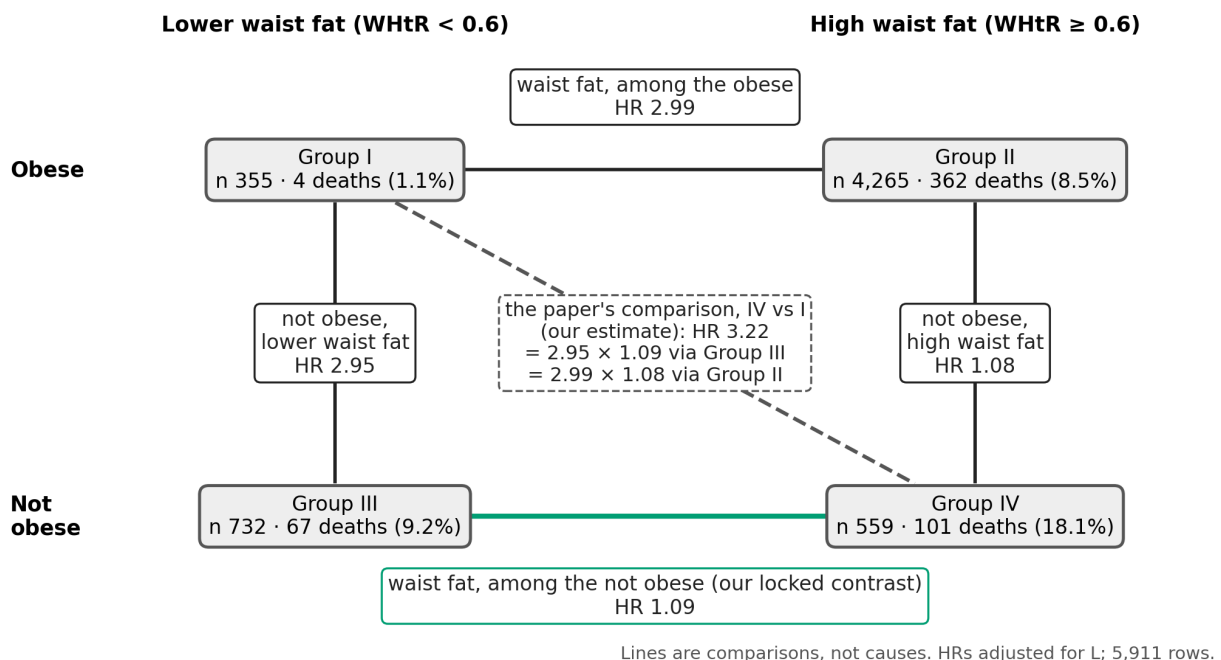
- `risk6()` sets everyone in a stratum to a group, predicts survival at 6 years, averages: obese 2.3% (I) and 6.4% (II); not obese 9.1% (III) and 9.8% (IV). Risk differences 4.1 and 0.7 points; difference 3.4. This is Week 4's g-computation with `survfit()` in place of `predict()`.

Slide 17 — Effect modification: waist fat, by obesity (4 min) · the summary of 16a–16c



- **The Week 4 link.** Week 4's Q2 model, waist fat adjusted for obesity and L with **no** product term, gave **1.24 (0.93–1.67)**: one HR for everyone. Let it differ by obesity:
- **Multiplicative.** **2.99 (1.11–8.09)** among the obese (II vs I), **1.09 (0.79–1.50)** among the not obese (IV vs III). Ratio, obese ÷ not obese, **2.74 (0.97–7.71)**: likelihood-ratio p = 0.03, Wald p = 0.06; profile-likelihood interval 1.10–9.18.
- **Additive.** Standardised six-year risks *within each stratum* (everyone in the stratum set to lower vs high waist fat): obese 2.3% → 6.4%; not obese 9.1% → 9.8%. Difference, obese – not obese, **+3.4 (–0.1 to +6.6)** points.
- **Verdict.** Both scales point the same way. With 4 deaths in the reference cell the likelihood-ratio test is the better guide than the Wald test, and it says p = 0.03; but the honest reading is “**borderline** — possibly larger among the obese, resting on four deaths”. Do not let either threshold decide.
- Bootstrap: 500 percentile resamples; **10** drew none of Group I's deaths. They are kept.

Slide 18 — The paper's IV vs I, taken apart (4 min)



- IV vs I changes **both** exposures. It splits two ways: 2.95×1.09 via Group III (not obese, then waist fat), or 2.99×1.08 via Group II (waist fat, then not obese). The product identity holds both ways; **the attribution does not** — because the two exposures interact. That is today’s topic in one picture.
- Either way, the big step is the one that **leaves Group I** and its 4 deaths. The one-exposure-at-a-time contrast is the locked **IV vs III: 1.09**, and that is the reason the course locked it.
- **3.22 is our estimate of the paper’s comparison**, adjusted for L. The paper reports 2.9 (its Model 2, aHR 2.9). With the paper’s Model 1 instead of L, P2’s locked estimate is **1.06 (0.77–1.45)** on P2’s 5,939 rows; the story is the same.

Slides 19–20 — Interaction: four cells, one reference (5 min) · one deck slide

- Knol–VanderWeele table, every cell vs Group I: II **2.99 (1.11–8.09)**, III **2.95 (1.07–8.13)**, IV **3.22 (1.17–8.87)**; within-row and within-column simple effects, including IV vs II, **1.08 (0.86–1.35)**. With Group I as reference the two “exposures” are not being obese (A) and high waist fat (M): $RR_{10} = III$, $RR_{01} = II$, $RR_{11} = IV$ — on the slide.
- Multiplicative: $HR_{IV} / (HR_{II} \times HR_{III}) = 0.37 (0.13–1.03)$ — the same product term as the forest plot’s 2.74, coded the other way; the slide says so.
- Additive: RERI **–1.72 (–4.57 to 1.13)**, AP -0.53 , S 0.56. Standardised six-year risks over everyone: **2.5%, 7.0%, 6.9%, 7.5%** $\rightarrow$ IC **–3.9 (–6.7 to –0.5)** points; the risk-based RERI ($IC \div R_{00}$) is -1.56 .
- **Verdict.** Point estimates less than additive — the same borderline evidence as before, coded the other way. Every comparison with Group I leans on its 4 deaths (1.09 and 1.08 do not). And it **cannot be read as causal**: U, obesity’s confounder, was never measured (Panel B). The RERI interval includes 0 and the IC interval does not; different measures; no threshold picks between them.

Slide 20a — In R: RERI, AP, S and IC for the four groups (3 min)

- The toy’s formulas on `exp(coef(fit_joint))` (all vs Group I): RERI -1.72 , AP -0.53 , S 0.56; and `risk6()` over all 5,911 people set to each group: 2.5, 7.0, 6.9, 7.5% $\rightarrow$ IC -3.9 points. The slide reads the risks in pairs: waist fat adds 4.5 points if everyone were obese and 0.6 if no one were; $0.6 - 4.5 = -3.9$.

Part 5 · Time-to-event tools, and a subgroup claim · 17 minutes

Slide 21 — The right tools for a time-to-event outcome (3 min)

- One table, two columns: **effect modification** (2.74; +3.4 points) and **interaction** (0.37; IC -3.9 points; RERI -1.72 , risk-based -1.56). Each question keeps its own coding.
- Multiplicative: Cox product term. Additive: standardised risks at a fixed time (Week 4’s g-computation with a survival curve; within strata for modification, over everyone for interaction). Shortcut: RERI from HRs, close to the risk-based RERI when deaths are uncommon over follow-up.
- **Not** “ever died” + logistic: that discards follow-up and censoring.
- A binary outcome with fixed follow-up (Thursday) is different: logistic is right. But ORs approximate RRs only when the outcome is rare — L4’s death outcome is about 40% — so an RERI from ORs there is not the RERI from risks.

Slide 22 — A dramatic subgroup claim (4 min)

- Crude IV vs I within each sex, 6,048 rows (P3’s code; Group I has 375 people there): men **36.14 (8.86–147.48)**, women **4.11 (0.99–17.12)**. Group I: **2 / 317** men (0.6%) and **2 / 58** women (3.4%) died.
- Crude ratio, women vs men: **0.11 (0.02–0.85)** — “significant”, for a contrast that changes both exposures, in a crude model, divided by four deaths.
- Say what each number rests on, not why it is large. No mechanism.

- **Provenance note.** The P3 exemplar’s prose table once printed 28.2 / 4.08 — the same model on the full file (6,371) — above code that fits the 6,048 and prints 36.14 / 4.11. It now prints the code’s numbers. The old Week 5 deck’s “men $\approx$ 28” came from that mismatch.

Slide 22a — In R: count Group I by sex first (2 min)

- `with(subset(dom, group == "I"), table(female, died = dead))`: 2 deaths in each sex.

Slide 23 — The same question, asked properly (4 min)

- Locked contrast IV vs III, sex as modifier, one test of the product term (likelihood-ratio p throughout): crude **0.62 (0.26–1.43)**, p 0.278 (P3, 1,296 rows); Model 1 **1.00 (0.42–2.35)**, p 0.993 (P3); pre-exposure set **1.13 (0.48–2.66)**, p 0.78 (1,291 rows).
- P3’s six-year RDs, in points with their intervals: +0.8 (–3.3 to +4.2) in men, +0.5 (–7.3 to +7.6) in women — small point estimates, wide intervals.
- The honest statement: no evidence of modification at this precision; the interval is compatible with the effect in women being less than half, or more than twice, that in men.
- **Withdrawn, and said once in the notes.** An earlier version of this lecture called the crude gap “confounding by baseline risk” and told students to report a common multiplicative effect. The data cannot say what produced the gap.

Slide 23a — In R: one test for sex, on the locked contrast (2 min)

- Restrict to Groups III and IV; `IV * female` with L; ratio 1.13 (0.48–2.66), likelihood-ratio p 0.78 (the third row of Slide 23’s table).

Slide 24 — Honest subgroup reporting (2 min)

- Pre-specify one modifier with a reason; say which question (EM or interaction); count every cell; both scales with intervals and a named reference; a big crude gap is not evidence of modification and a null test is not evidence of none; no mechanism without evidence.
- This is the P3 checklist.

Close · 4 minutes

Slide 25 — Key takeaways; next week (4 min)

- Six takeaways (deck). **Next:** Week 6, survey weights; design-aware, P3’s Model 1 sex ratio (1.00) becomes **2.53 (0.92–6.98)** (M1).
- **What today cannot fix:** unmeasured U; selection on $FLI \geq 60$, computed from both of today’s exposures, which need not bias every stratum equally and so can create or hide a difference between strata. M1 asks groups to rank these threats from their own evidence.
- Exit question: *one modifier for your own paper — and why it is fixed before your exposure.*

Slide 26 — Where to read more; references; dagitty appendix (uncounted)

Thursday — L4 (EpiMethods confoundingE2) · for the TA and the instructor, not in the deck

The Week 4 deck carries no lab bridge, and neither does this one. The L4 handout carries the short version of these notes; they are worth five minutes at the start of Thursday’s session.

- **Data.** The RHC data again (Week 4's L3): the 1,439 complete cases, restricted to $\text{DASI} > 20 \rightarrow 623$ patients. Outcome: the binary death variable, **40%** died. Exposure: `no_rhc_status`; modifier / second exposure: `no_dnr_status`.
- **The exercise's reference cell.** The exercise text says it changes the reference levels "to No RHC and No"; its code does the reverse. Both indicators are coded 1 for "No", so the reference cell (0, 0) is **RHC and a DNR order** — the lowest-risk cell, as Slide 11 says RERI needs.
- **Count first — that reference cell holds 3 patients.** Only **23** of the 623 have a DNR order (20 without RHC, 3 with). Run with `(high_dasi_data, table(rhc_status, dnr_status, death_status))`: the RHC + DNR cell has 3 patients and 1 death. Everything in Problem 2 divides by it:
 - OR for no RHC among DNR patients **29.8 (1.14–781)**;
 - RERI **–28.5 (–125 to 68)**, AP **–14.0**, S 0.04 (0.002–0.80);
 - product term $p = 0.056$ (Problem 1's $p = 0.048$, with RHC's confounders only).
 - The only well-supported stratum estimate is no RHC among patients **without** a DNR order: OR **1.21 (0.78–1.86)**, 600 patients. This is Slide 15's lesson in the lab's data: Group I has 4 deaths; the lab's reference cell has 3 patients.
- **Read the intervals, not interactionR's p column.** `interactionR` (0.1.7) computes the RERI, AP and S p-values one-sided or on the wrong scale: S prints $p = 0.98$ while its own interval (0.002–0.80) excludes 1. The exercise tells students to compare those p-values with 0.05. S's interval excludes 1 while RERI's includes 0: different measures on one 3-patient cell — Slide 19's lesson again.
- **interactionR prints the wrong way round in Problem 1.** With `em = TRUE` it treats the **first** name in `exposure_names` as the modifier, so as called it shows DNR's effect by RHC — the reverse of Problem 1's question (Slide 9). Read Problem 1 from the Publish rows (no RHC vs RHC: **33.0 (1.79–609)** with a DNR order, **1.22 (0.80–1.85)** without), or swap the order of `exposure_names`.
- **Statements in the exercise text to correct out loud:**
 1. "SI = 1: no multiplicative interaction" — S is an **additive** measure (Slide 11).
 2. "RERI < 0 ... the exposures may be protective when combined" — a negative RERI means **less than additive**, not protective.
 3. Interaction as "a departure from multiplicativity" — differs from the *Concepts* page and from Tuesday (the joint effect of two exposures, on either scale).
- **Death is common (40%),** so ORs overstate RRs and the OR-based RERI is not the risk-based RERI (Slide 21).
- **The 1,439 are complete cases,** which Week 4 showed is selection on early survival (the ADL score). It carries into L4 unchanged.
- **DNR as the modifier fails Tuesday's own rule unless it came first.** DNR and RHC are both day-1 decisions, and which came first is not recorded. If a DNR order could follow catheterisation, DNR is a consequence of the exposure, and stratifying on it opens $\text{RHC} \rightarrow \text{DNR} \leftarrow \text{severity} \rightarrow \text{death}$. Ask the room whether a DNR order could follow catheterisation, and read Problem 1 as conditional on the answer. The same doubt complicates the interaction reading in Problem 2.
- To reproduce: the numbers above come from running the solution's two `glm()` calls on `rhc_data.rds` from `confoundingEx2.zip`, plus `interactionR::interactionR()` and `Publish::publish()` as the solution calls them.

If you are running behind

Drop or shrink in this order. The first two cost the least.

1. **The toy "In R" slides (10a, 12a) — 5 $\rightarrow$ 1 min.** Show the output, skip the code; the MASLD "In R" slides carry the same formulas. **Slide 12 — same data, two answers (3 $\rightarrow$ 1 min).** Say the rule (harmful exposures, no multiplicative interaction positive additive) and move on.
2. **Slides 22–22a — the dramatic subgroup claim (6 $\rightarrow$ 3 min).** Show the table; say "count first".
3. **Slides 16a–16b (5 $\rightarrow$ 2 min).** Keep the flip trick and the likelihood-ratio p.
4. **Slide 9 — which way round (3 $\rightarrow$ 1 min).** Go straight to 9a; never drop the rule itself.

Never drop Slides 8 (both deck slides), 9a, 15, 16c, 17 or 18. 9a is the rule students apply in P3.

The two-question DAG and its assumptions, the cell counts, the effect-modification result and the IV-vs-I decomposition are the argument; everything else supports one of the four.

Reading map — each segment and where students can review it

Segment	EpiMethods / course page	Key reference
Definitions (Slides 4–6)	confounding0 <i>Concepts</i> , “Formal Definitions”; effect-modification video and slides	VanderWeele 2009; Bours 2021; Karim ch. 8 §8.2–8.5
Adjustment sets (Slides 7–9)	confounding0 , “Implications for Confounding Control”	Karim ch. 8, Fig. 8.1
Scale (Slides 10–12)	confounding0 , “The Role of the Scale”; confounding5 (Week 4)	VanderWeele & Knol 2014
RERI, AP, S; reference cell (Slides 11, 15, 19–20)	confounding9 , “Additive interaction measures”; interaction tutorial	Knol et al. 2011; Hosmer & Lemeshow 1992
Joint variable vs product term (Slide 16)	confounding9	Westreich & Greenland 2013
Reporting (Slides 13, 24)	confounding0 , “Reporting guideline”	Knol & VanderWeele 2012; Karim ch. 8 §8.7
Thursday	confoundingE2 , confoundingE2solution	—

Three things to know before assigning the reading.

- **Karim, ch. 8.** Sections 8.1–8.5 and 8.7 match today; §8.6 (propensity-score methods for subgroup effects) belongs after the propensity-score weeks. Three slips in the assigned sections: §8.4.3.1 gives the ratio of odds ratios as 1.27 “with a 95% confidence interval of 1.04 to 1.27” (Table 8.3: 1.04–1.56); the same paragraph calls $OR_{M=0} = 2.93$ “the low-income group” although §8.3 codes high income as $M = 0$; and §8.5.1 describes AP as the share “attributed to the exposure” rather than to the interaction.
- **The interaction tutorial** states that RERI “ranges from 0 to ∞ ” and AP “from -1 to 1 ”. Both can be negative, and AP can fall below -1 (today’s RERI is -1.72 ; the lab’s AP is -14.0). Say so if students are pointed at that page.
- **The EpiMethods lab text** — see *Thursday* above.

Appendix — the two questions as dagitty code

```
dag {
  Waist [exposure]  Death [outcome]  U [latent]
  L -> Obese  L -> Waist  L -> Death
  U -> Obese  U -> Death
  Obese -> Waist  Obese -> Death  Waist -> Death
}
```

With **Waist** as the only exposure dagitty returns $\{L, Obese\}$. Add **Obese** as a second exposure and no set exists; remove the latent tag from **U** and $\{L, U\}$ appears. Each of the second Slide 8’s assumptions, broken, removes Panel A’s set: reverse **Obese** $\rightarrow$ **Waist**; add **U** $\rightarrow$ **Waist**; add a latent **G** $\rightarrow$ **Obese**, **G** $\rightarrow$ **Waist**; add a latent income node; or add **FLI** [adjusted] **Waist** $\rightarrow$ **FLI** **Waist** $\rightarrow$ **TG** **TG** $\rightarrow$ **FLI** **TG** $\rightarrow$ **Death**.

References

- Banack HR, Kaufman JS. The obesity paradox: understanding the effect of obesity on mortality among individuals with cardiovascular disease. *Prev Med* 2014;62:96–102.

- Bours MJL. Tutorial: a nontechnical explanation of the counterfactual definition of effect modification and interaction. *J Clin Epidemiol* 2021;134:113–124.
- Hosmer DW, Lemeshow S. Confidence interval estimation of interaction. *Epidemiology* 1992;3:452–456.
- Karim ME. Effect modification in non-randomized studies: methods and applications with propensity scores. Chapter 8 (book chapter).
- Lajous M, Bijon A, Fagherazzi G, et al. Body mass index, diabetes, and mortality in French women: explaining away a “paradox”. *Epidemiology* 2014;25:10–14.
- Knol MJ, VanderWeele TJ. Recommendations for presenting analyses of effect modification and interaction. *Int J Epidemiol* 2012;41:514–520.
- Knol MJ, VanderWeele TJ, Groenwold RHH, et al. Estimating measures of interaction on an additive scale for preventive exposures. *Eur J Epidemiol* 2011;26:433–438.
- Kueh MTW, et al. Body weight categories and fat distribution in relation to all-cause mortality among adults with metabolic dysfunction-associated steatotic liver disease. *BMJ Open* 2026;16:e113719.
- VanderWeele TJ. On the distinction between interaction and effect modification. *Epidemiology* 2009;20:863–871.
- VanderWeele TJ, Knol MJ. A tutorial on interaction. *Epidemiol Methods* 2014;3:33–72.
- Westreich D, Greenland S. The Table 2 fallacy: presenting and interpreting confounder and modifier coefficients. *Am J Epidemiol* 2013;177:292–298.